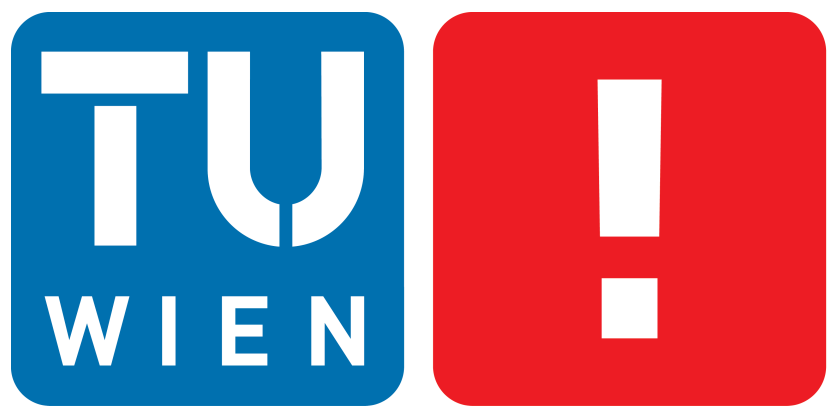




Open Data Quality

Assessment and Evolution of (Meta-)Data Quality in the Open Data Landscape Sebastian Neumaier

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What is Open Data?

The definition of **Open Data** [1] can be summed up in the following statement:

***Freely available data,
published in a machine readable format
which allows everybody
to do everything without restrictions
at anytime.***

Open Data is mainly published on **Open Data Portals**:

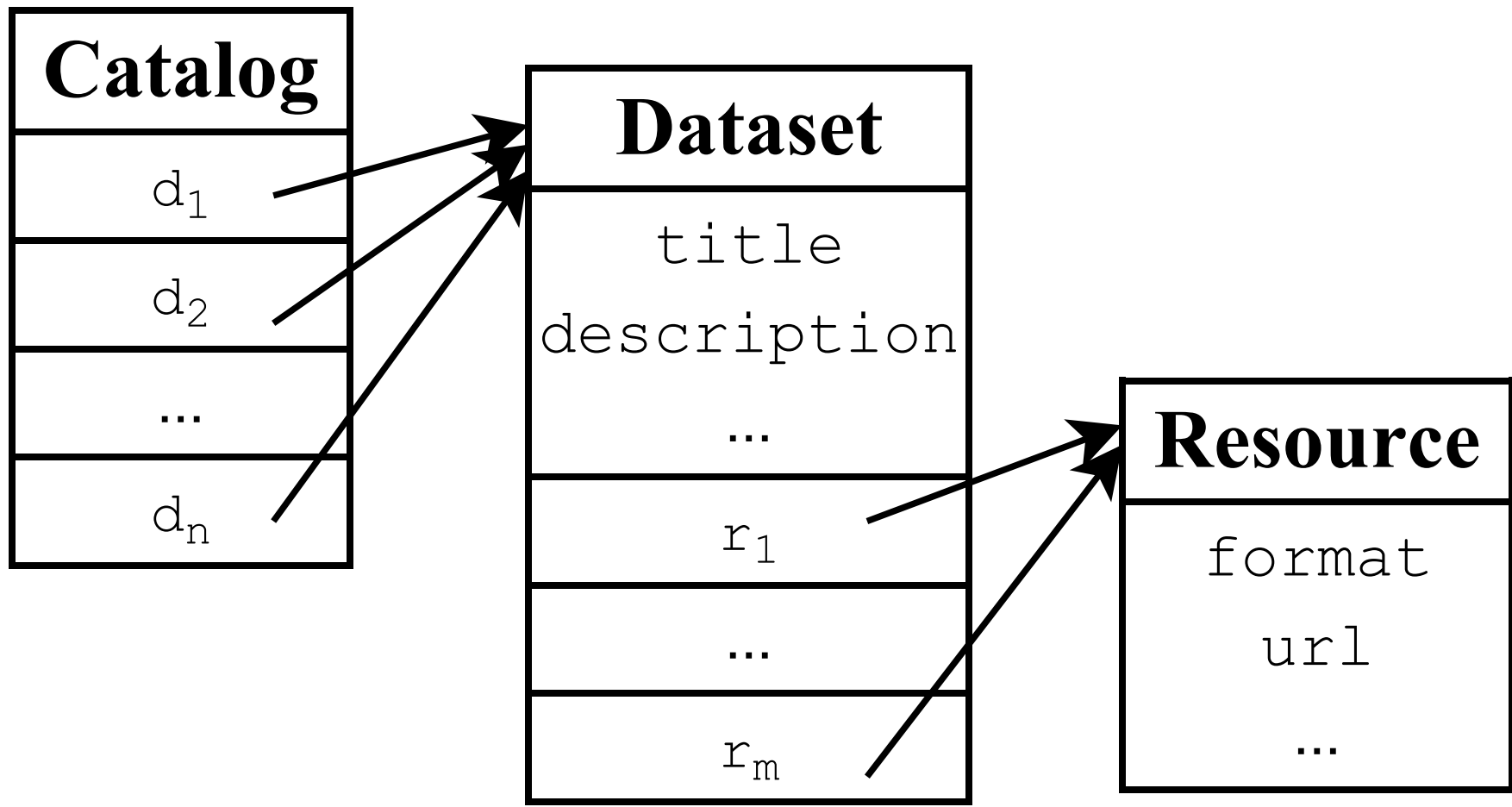


Figure: High Level Structure of an Open Data Portal.

- **Datasets** hold **metadata** (i.e., basic information) of resources, e.g. authorship or licensing information.
- **Resources** are available for access or download in one or more formats, e.g., CSV, PDF and Microsoft Excel.
- Data portals offer an **API** to retrieve datasets.

Motivation:

Poor quality and **high heterogeneity** of metadata and data sources in Open Data portals is a serious risk that could disrupt the success of the Open Data movement.

Problem Statement:

There exist no objective and **comprehensive reports** about the actual quality and **evolution** of portals.

Findings

Usage and Completeness:

- Extra keys (green) partially unused: Bottom left portals have low values
- Indication that there are redundant metadata keys

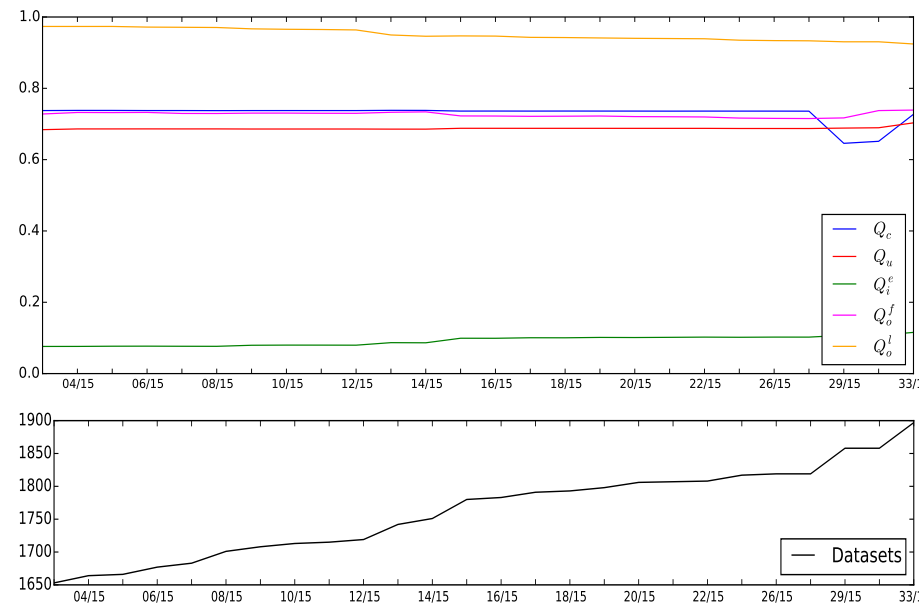


Figure: Ex.: evolution on data.gv.at.

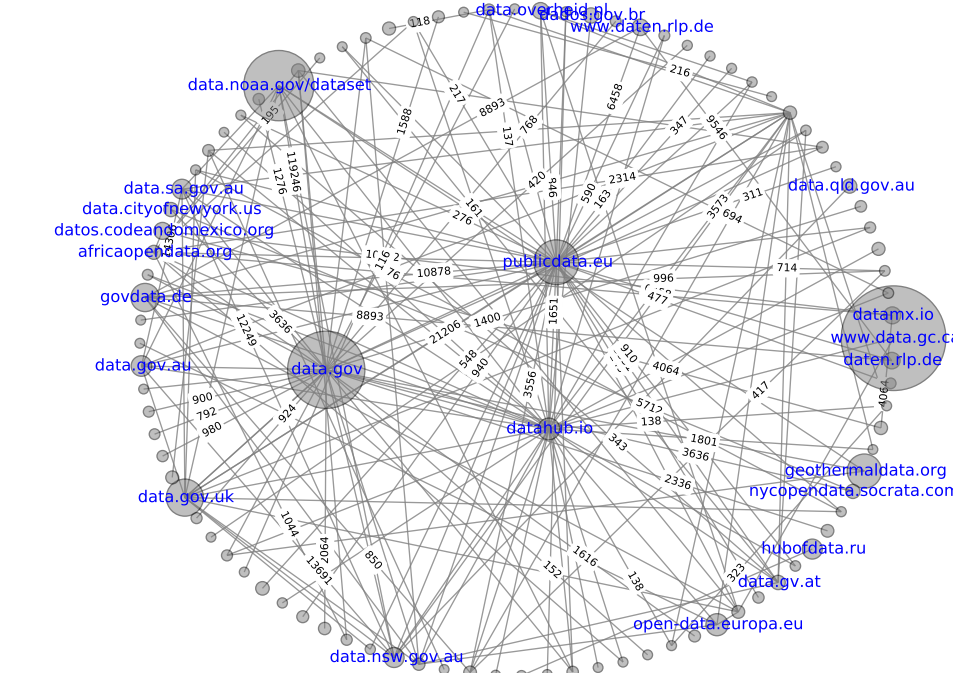


Figure: Shared Resources.

Format	%	p
PDF	25.6	89
HTML	23	70
other	6.9	6
CSV	5.8	107

Table: Top-4 formats.

Parts of the findings have been published in [3] and [4].

Research Questions & Approach

Research Questions:

- Does Open Data follow the theoretical definition?
- Can we confirm the quality and heterogeneity issues in Open Data?
- Are we able to assess the development and growth-rate of Open Data?

Approach:

- Objective **quality metrics** that can be calculated in an automated way.
- A **monitoring framework** which regularly monitors data catalogs.
- Report **findings** based on monitoring a corpus of over 120 CKAN data portals.

Quality Metrics & Automated Assessment Framework

Quality Metrics:

	Dimension	The extent to which ...
Q_r	Retrievability	... metadata and resources can be retrieved.
Q_u	Usage	... available metadata keys are used to describe a dataset.
Q_c	Completeness	... the used metadata keys are non-empty.
Q_a	Accuracy	... certain metadata values accurately describe resources.
Q_o	Openness	... licenses and file formats conform to the open definition.
Q_i	Contactability	... the data publisher provide contact information.

Automated Assessment Framework:

We developed the monitoring framework "*Open Data Portal Watch*", similarly to [2], retrieving datasets from the CKAN, Socrata and OpenDataSoft software.

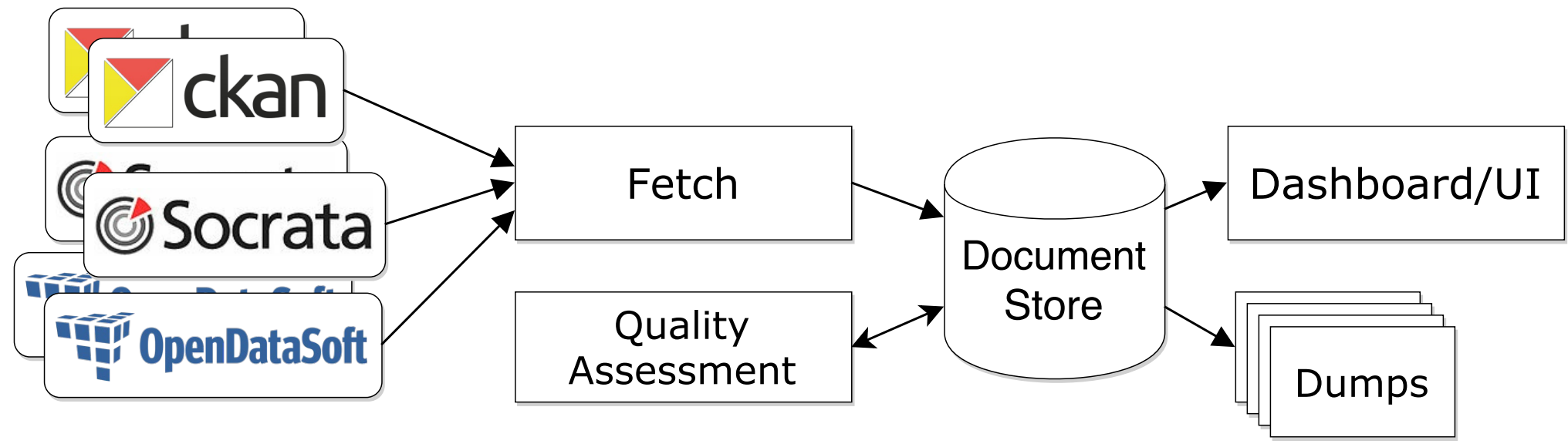


Figure: The *Open Data Portal Watch* architecture.

"<http://data.wu.ac.at/portalwatch/>

Fetch: Retrieves data weekly and stores it in a DB.

QA: Metrics are assessed by the QA component.

UI: Findings are reported on a dashboard component."

Dumps: Raw metadata is freely available on UI component.

Further Work

Metadata homogenization: Mapping of different metadata formats to integrate other portal software.

Automated Quality Improvement: Use accurate data for automated correction.

References

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- Jürgen Umbrich, Sebastian Neumaier, and Axel Polleres. Quality assessment & evolution of open data portals. In *IEEE International Conference on Open and Big Data. Rome, Italy (August 2015)*, <http://www.polleres.net/publications/umbr-et-al-2015OBD.pdf>, 2015. Best paper award.
- Jürgen Umbrich, Sebastian Neumaier, and Axel Polleres. Towards assessing the quality evolution of open data portals. In *ODQ2015: Open Data Quality: from Theory to Practice Workshop. Munich, Germany (Mar 2015)*, <http://polleres.net/publications/umbr-et-al-2015ODQ.pdf>, 2015.

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